INSTITUTIONAL SYSTEM SPECIFICATION & OPERATIONAL MANUAL

Delta Exchange Multi-Agent Trading System

A Multi-Agent Architecture for Automated Market Regime Analysis, Quantitative Risk Sizing, Drawdown Circuit Breakers, and Leveraged Bracket Order Execution via the Model Context Protocol (MCP).

ARCHITECTURE
5-Agent Collaborative Pipeline

PROTOCOL
FastMCP & Native JSON-RPC 2.0

EXCHANGE
Delta Exchange (India / Global / Testnet)

GOVERNANCE
Strict 1% Account Risk & 2:1 R:R

1 Executive Summary & Core Engineering Tenets

The Delta Exchange Multi-Agent Trading System transforms retail and algorithmic cryptocurrency trading by implementing an institutional-grade division of labor. Inspired by the principles of `tradermonty/claude-trading-skills`, the system establishes a zero-trust execution environment where trade generation and trade execution are strictly segregated by a mandatory, non-negotiable **Pre-Trade Discipline Gate**.

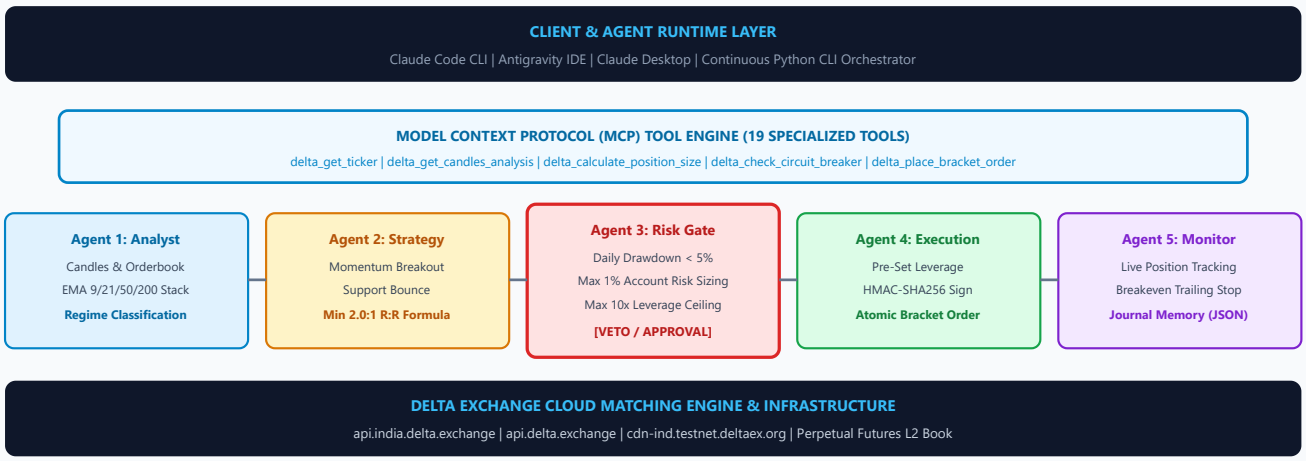
> **The Golden Rule of Capital Preservation**
> No market order or limit order can be submitted directly to Delta Exchange by an analytical model. A trade candidate is merely a hypothesis until it survives the mathematical veto filters of the Risk Gatekeeper. If daily drawdown breaches 5.0%, if margin exceeds available collateral, or if the Reward-to-Risk ratio is under 2.0:1, the trade is dropped immediately.

High-Level Architectural Components

Component Layer	Underlying Module	Primary Responsibility	Integration Standard
1. Protocol Engine	`src/delta_mcp_server.py`	Exposes 19 high-level tools to AI agents and IDEs (Claude Code, Antigravity, Cursor).	Model Context Protocol (FastMCP / Stdio JSON-RPC)
2. Client & Auth Engine	`src/delta_client.py`	HMAC-SHA256 authenticated REST interface to Delta API v2 endpoints with connection caching.	Delta Exchange REST v2 Specification
3. Quantitative Engine	`src/technical_analysis.py`	Pure vectorized technical indicator suite: EMA (9/21/50/200), RSI-14, ATR-14, MACD, Support/Resistance.	Mathematical Candlestick Modeling
4. Risk Engine	`src/risk_manager.py`	Account equity drawdown circuit breakers, dynamic 1% risk position sizing, leverage caps.	Persistent Circuit Breaker State Machine
5. Multi-Agent Team	`multiagent_team/*.py`	5 specialized agents collaborating sequentially to analyze, vet, execute, and journal trades.	Autonomous Pipeline Orchestrator

2 Complete System Architecture & Data Flow

The system interfaces between external AI runtime environments (Claude Code CLI, Claude Desktop, Antigravity IDE) and Delta Exchange's core order matching engine through a layered gateway.



Authentication & The HMAC-SHA256 Signing Protocol

Security is maintained through strict cryptographic request signing. All private HTTP calls (balances, positions, orders) require a unique epoch timestamp and an HMAC-SHA256 signature generated using the secret key:

```
Signature Payload = Method + Timestamp + Path + QueryString + Body
Signature = HMAC_SHA256(Key=DELTA_API_SECRET, Message=Signature Payload).hex()
```

This signature is transmitted via the `signature` header along with `api-key` and `timestamp`. Replay attacks are rejected by Delta Exchange because timestamps with a clock skew exceeding 30 seconds are rejected with an authentication failure.

3 The 5-Agent Pipeline: Detailed Specifications

Each agent functions as an independent state machine with explicit preconditions, mathematical responsibilities, and immutable guardrails.

Agent 1: Market Regime & Technical Analyst (`regime_analyst.py`) — ANALYTICAL STAGE

Objective: Formulate an objective, bias-free quantitative model of market structure across timeframes.
Inputs: 100 historical candles at 1-hour (`60`) and 4-hour (`240`) resolution, Level 2 orderbook depth.
Indicators Calculated: - *EMA Ribbon:* 9, 21, 50, 200 periods.
- *Momentum:* RSI-14 (Wilder Smoothed) and MACD (12, 26, 9) signal line.
- *Volatility:* ATR-14 (Average True Range).
- *Structural Pivots:* 20-period swing highs (resistance) and swing lows (support).
Non-Negotiable Rule: The Regime Analyst is strictly read-only. It is architecturally prohibited from emitting trade orders or suggesting entry sizing.

Agent 2: Strategy & Setup Generator (`strategy_agent.py`) — ALPHA GENERATION

Objective: Translate market regimes into high-expectancy trade proposals with explicit risk boundaries.
Strategy Playbooks:
1. *Trend Continuation Long:* Triggered when Regime is `STRONG_BULLISH` and RSI is healthy ($50.0 \le \text{RSI} \le 72.0$). Stop-Loss is placed below EMA 21 or $1.5 \times \text{ATR}_{14}$.
2. *Trend Continuation Short:* Triggered when Regime is `STRONG_BEARISH` and RSI is weak ($28.0 \le \text{RSI} \le 50.0$). Stop-Loss is placed above EMA 21 or $1.5 \times \text{ATR}_{14}$.
3. *Mean Reversion Bounce:* Triggered when price tests key structural swing support with oversold RSI (< 32.0).
Non-Negotiable Rule: Every trade proposal must have a mathematically validated Reward-to-Risk ratio $\ge 2.0:1$.

Agent 3: Risk & Discipline Gatekeeper (`risk_gatekeeper.py`) — SAFETY VALVE

Objective: Capital preservation. Validates account safety metrics and calculates fractional position size.
Discipline Filters:
1. *Account Circuit Breaker:* Daily drawdown must be $< 5.0\%$. Peak drawdown must be $< 10.0\%$.
2. *Exposure Ceiling:* Maximum 3 concurrent open positions across all instruments.
3. *Position Sizing Math:* Max dollar loss at stop-loss must not exceed 1.0% of available equity.
4. *Margin Feasibility:* Required margin must be $\le 40\%$ of available collateral to avoid over-leveraging.
Non-Negotiable Rule: Agent 3 has unilateral veto power. If rejected, execution halts immediately.

Agent 4: Execution & Order Manager (`execution_agent.py`) — ORDER EXECUTION

Objective: Reliable, low-slippage order execution on Delta Exchange.
Execution Sequence: Sets leverage $\le 10\times$ on the instrument $\rightarrow$ formats atomic bracket order payload $\rightarrow$ submits primary entry leg with attached Stop-Loss and Take-Profit child orders $\rightarrow$ captures Delta Order ID.
Non-Negotiable Rule: Never submit naked orders. Entry without simultaneous Stop-Loss is forbidden.

Agent 5: Portfolio Monitor & Memory Journaler (`portfolio_monitor.py`) — MEMORY & LIFECYCLE

Objective: Real-time position tracking, trailing stop management, and persistent trade memory.
Post-Fill Protocols: Monitors mark price vs entry $\rightarrow$ when position achieves $> 1.0\times$ risk in profit, automatically adjusts Stop-Loss order to breakeven $\rightarrow$ logs full metadata to `state/trade_journal.json` $\rightarrow$ generates daily markdown digests in `reports/` .

4 Quantitative Foundations & Mathematical Formulations

The quantitative model operates without subjective human bias, using strict statistical criteria to determine regime state and sizing.

1. Multi-Timeframe Trend Structure (The EMA Ribbon)

Exponential Moving Averages give higher weighting to recent price actions. The system calculates four periods:

```
Multiplier (k) = 2 / (Period + 1)
EMA_today = (Price_today * k) + (EMA_yesterday * (1 - k))
```

Market Regime Classification	Moving Average Alignment	RSI-14 Requirement	Permitted Trading Action
STRONG_BULLISH	$\text{Price} > \text{EMA}_9 > \text{EMA}_{21} > \text{EMA}_{50} > \text{EMA}_{200}$	$50.0 \le \text{RSI} \le 72.0$	Long Trend Continuation Only
MILD_BULLISH	$\text{Price} > \text{EMA}_{21} > \text{EMA}_{50}$ (Mixed Stack)	$45.0 \le \text{RSI} \le 60.0$	Long Pullback / Support Bounce
NEUTRAL_RANGEBOUND	Price oscillating between $\text{EMA}_9 \text{ and } \text{EMA}_{50}$	$40.0 \le \text{RSI} \le 60.0$	**NO TRADING** (Preserve Capital)
MILD_BEARISH	$\text{Price} < \text{EMA}_{21} < \text{EMA}_{50}$	$40.0 \le \text{RSI} \le 55.0$	Short Resistance Rejection
STRONG_BEARISH	$\text{Price} < \text{EMA}_9 < \text{EMA}_{21} < \text{EMA}_{50} < \text{EMA}_{200}$	$28.0 \le \text{RSI} \le 50.0$	Short Trend Continuation Only

2. Volatility Normalization: Average True Range (ATR-14)

Fixed dollar stops fail across changing market volatility. The system uses ATR-14 to expand stops during high-volatility regimes and contract them during low-volatility regimes:

```
TR_t = \max \big( High_t - Low_t, \; |High_t - Close_{t-1}|, \; |Low_t - Close_{t-1}| \big)
ATR_{14} = \frac{1}{14} \sum_{i=0}^{13} TR_{t-i}
```

3. The 1% Account Risk Position Sizing Engine

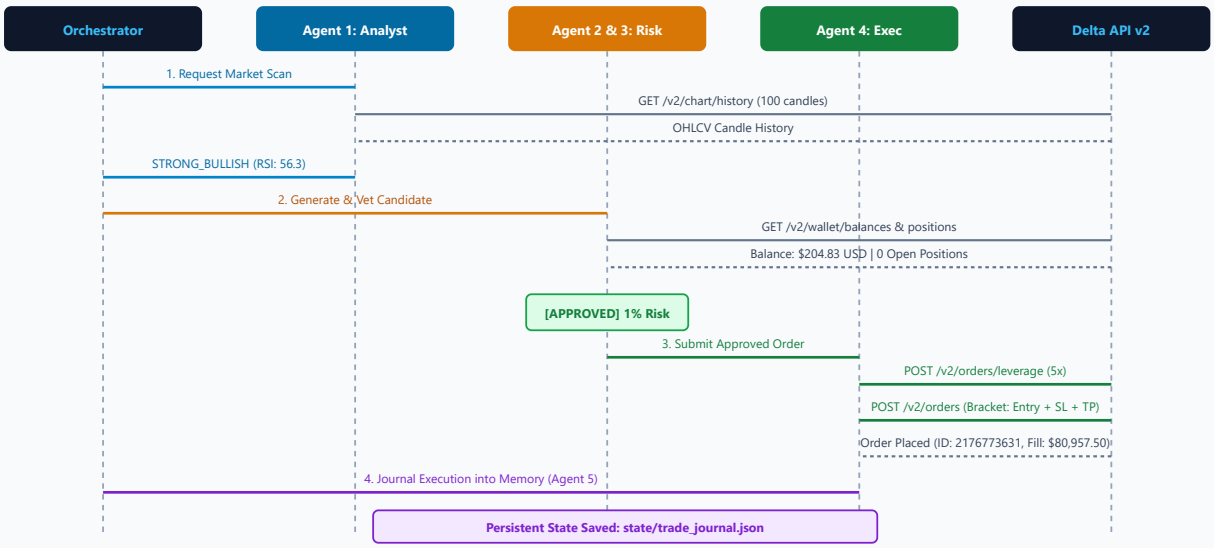
To ensure mathematical survival over hundreds of trades, the maximum capital risked on any individual setup is strictly capped at 1% of account equity:

```
\text{Dollar Risk Target (\$) } = \text{Account Equity} \times \text{MAX\_RISK\_PER\_TRADE\_PCT} \quad (0.01)
\text{Distance to Invalidation} = |\text{Entry Price} - \text{Stop Loss Price}|
\text{Calculated Contracts} = \left\lfloor \frac{\text{Dollar Risk Target}}{\text{Distance to Invalidation} \times \text{Contract Value}} \right\rfloor
```

If the calculated contract size is smaller than the minimum tradable unit (e.g. 1 contract), or if the required initial margin exceeds available balance, Agent 3 rejects the trade.

5 Complete Trade Execution Sequence

The end-to-end execution follows a rigorous chronological sequence of API interactions between the Multi-Agent Orchestrator and the Delta Exchange matching engine.



Bracket Order Anatomy

Delta Exchange allows linking child orders to a parent order. When the parent fills, the matching engine places the attached bracket stop-loss and take-profit orders in a pending state:

Leg Order	Order Type	Side	Reduce Only	Matching Engine Behavior
Primary Entry	`market_order`	BUY	False	Executes immediately against resting book liquidity.
Take-Profit Leg	`limit_order`	SELL	**True**	Sits in pending book at target price. Triggers when mark price $\ge \$81,760.10$.
Stop-Loss Leg	`stop_loss_order`	SELL	**True**	Triggers if mark price drops to stop price. Cancels TP leg automatically.

6 Live Execution Case Study: BTCUSD Perpetual

To verify system operation, the full multi-agent pipeline was executed live on Delta Exchange Testnet India (`https://cdn-ind.testnet.deltaex.org`). Below are the exact execution logs and matching engine timestamps.

Step 1: Multi-Agent Regime Analysis & Gatekeeper Approval

LIVE LOG

```
22:42:57 [INFO] RegimeAnalyst: Fetching market structure for BTCUSDT (res: 60)...
   Overall Market Posture (BTC Anchor): STRONG_BULLISH
   • BTCUSDT  Price: $81061.00  | RSI: 55.7  | Regime: STRONG_BULLISH
   • ETHUSDT  Price: $2351.95   | RSI: 56.9  | Regime: MILD_BULLISH
   Found prospective setup: BTCUSDT BUY | Strategy: Trend_Continuation_Long | R:R: 2.0:1
🛡️ [Agent 3: Risk Gatekeeper] Running Pre-Trade Discipline Gates...
   Available USD Collateral: $204.83 USD
   Max Risk per Trade (1%): $2.04 USD
   ✅ [APPROVED] BTCUSDT: 1 contract | Risk: $0.80 (0.39%) | Margin: $16.19 USD
```

Step 2: Matching Engine Execution & Order Fills

LIVE FILL

```
🚀 [Agent 4: Execution Agent] Setting leverage to 5x on Product ID 84 (BTCUSD)...
   Submitting bracket order: BUY 1 contract BTCUSD @ $80,957.50
   [SUCCESS] Primary Order ID: 2176773631 | State: FILLED
   [ATTACHED] Take-Profit Order ID: 2176773633 | Trigger: $81,760.10 | State: PENDING
   [ATTACHED] Initial Stop-Loss ID: 2176773632 | Trigger: $79,355.40 | State: PENDING
   Liquidation Price: $64,968.39 (20% safe buffer below mark price)
```

Step 3: Trailing Stop Adjustment to Breakeven (Zero-Risk Protocol)

RISK ELIMINATION

```
📊 [Agent 5: Portfolio Monitor] Mark price expanded to $81,096.92 (+0.17% unrealized profit).
   Executing Breakeven Protocol:
   1. Cancel old Stop-Loss Order ID 2176773632 ($79,355.40) -> [CANCELLED]
   2. Place new Breakeven Stop-Loss Order ID 2176774413 ($80,957.50) -> [ACCEPTED]
   STATUS: Trade is now a Free-Roll. Downside Risk: $0.00 | Profit Target: +$802.60 / BTC
```

Journal Memory Record (`state/trade_journal.json`)


```
{
  "trade_id": "trade_e898765d",
  "symbol": "BTCUSDT",
  "product_id": 84,
  "side": "buy",
  "contracts": 1,
  "entry_price": 80957.5,
  "stop_loss_price": 80957.5,
  "take_profit_price": 81760.1,
  "planned_rr_ratio": 2.0,
  "strategy_name": "Trend_Continuation_Long",
  "thesis": "Strong bullish EMA stack and positive RSI (56.3). Targeting momentum expansion.",
  "status": "OPEN",
  "entry_time": "2026-10-29T17:15:12.865Z",
  "order_ids": {
    "entry": 2176773631,
    "take_profit": 2176773633,
    "breakeven_stop": 2176774413
  }
}
```

7 Operational Runbook & Production Deployment

The trading system supports multiple operational modes ranging from interactive CLI verification to fully autonomous cron automation and Model Context Protocol IDE integration.

Execution Mode	CLI Command	Ideal Use Case	Interactivity
Connectivity Test	`python test_delta_connection.py`	Verifies API keys, wallet balances, and exchange endpoints.	Non-interactive diagnostic
Interactive Scan	`python main.py interactive`	Scans markets, calculates sizing, prompts user for confirmation.	Human-in-the-loop (`[y/N]`)
Continuous Autonomous	`python main.py auto`	Runs continuous 5-minute scan cycles; executes approved trades.	Fully autonomous
FastMCP Server	`python main.py mcp`	Powers Claude Desktop, Antigravity, and Claude Code tools.	AI Agent Driven (Stdio RPC)

Environment Configuration Reference (`.env`)

```
# Choose: "india" (live INR/USD), "global" (live USD), or "testnet_india" / "testnet"
DELTA_ENVIRONMENT=testnet_india
DELTA_BASE_URL=https://cdn-ind.testnet.deltaex.org
DELTA_API_KEY=your_delta_api_key_here
DELTA_API_SECRET=your_delta_api_secret_here

# Risk Parameters
MAX_RISK_PER_TRADE_PCT=0.01       # 1% equity risk per trade
MAX_CONCURRENT_POSITIONS=3         # Max 3 open positions simultaneously
MAX_DAILY_DRAWDOWN_PCT=0.05        # 5% daily circuit breaker
MAX_ALLOWED_LEVERAGE=10            # 10x max leverage ceiling
MIN_REWARD_TO_RISK_RATIO=2.0       # Minimum 2:1 R:R

# Watchlist & Mode
DEFAULT_SYMBOLS=BTCUSDT,ETHUSDT,SOLUSDT,XRPUSDT
TRADING_MODE=interactive
```

Emergency Circuit Breaker & Kill Switch Protocol

> 🚨 **Emergency Procedure: Instant Position & Order Liquidation**
>
> If high-impact macroeconomic events or exchange outages threaten account stability, you can immediately cancel all pending orders and close all open positions using the built-in MCP commands:

```
python -c "from src.delta_client import DeltaExchangeClient; c = DeltaExchangeClient();
c.cancel_all_orders(); c.close_position('BTCUSDT')"
```